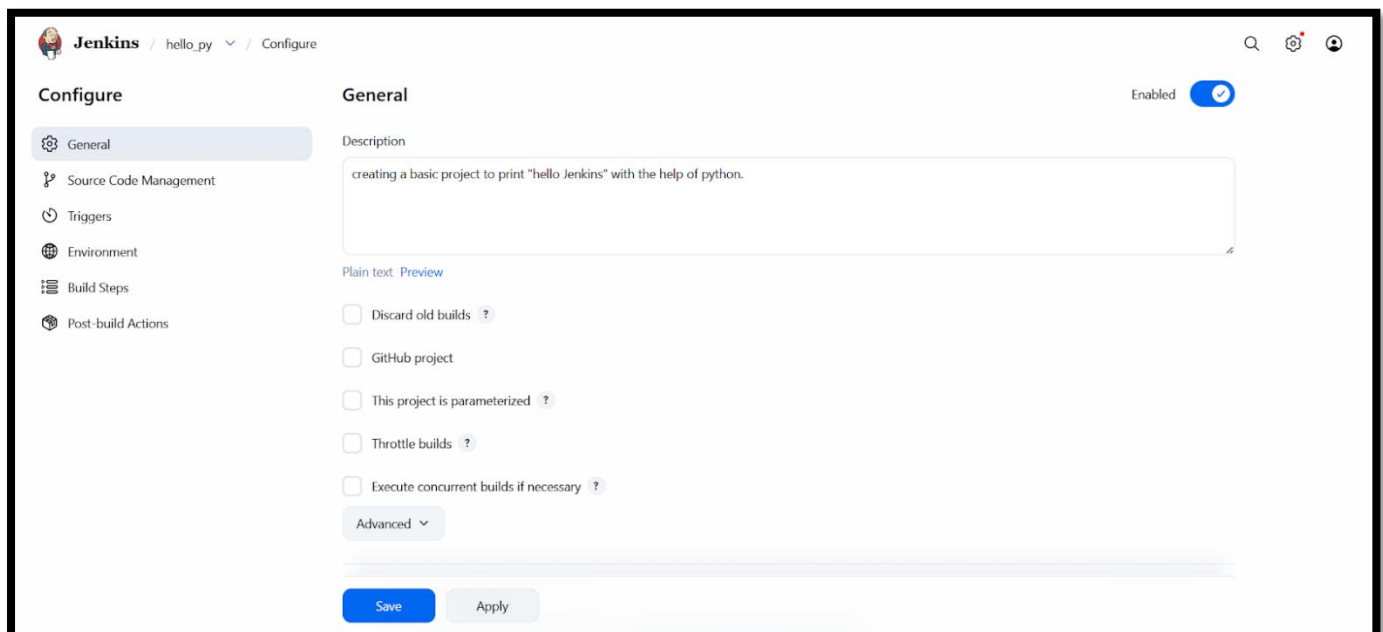
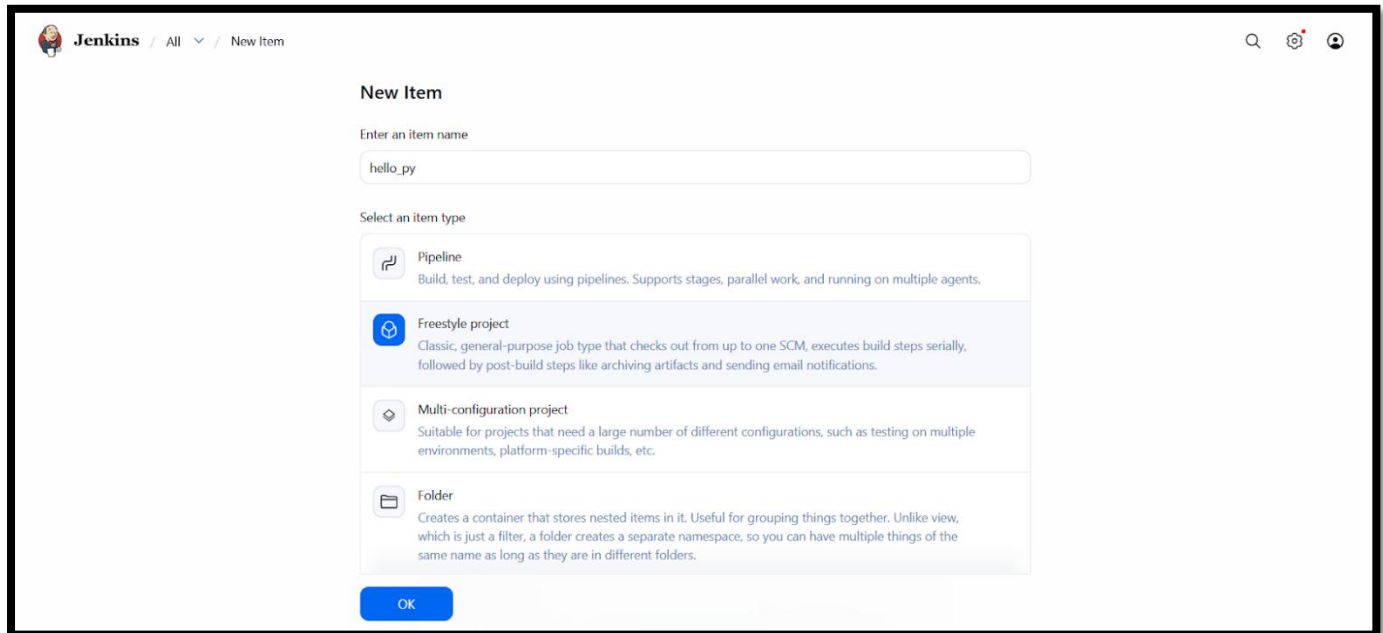


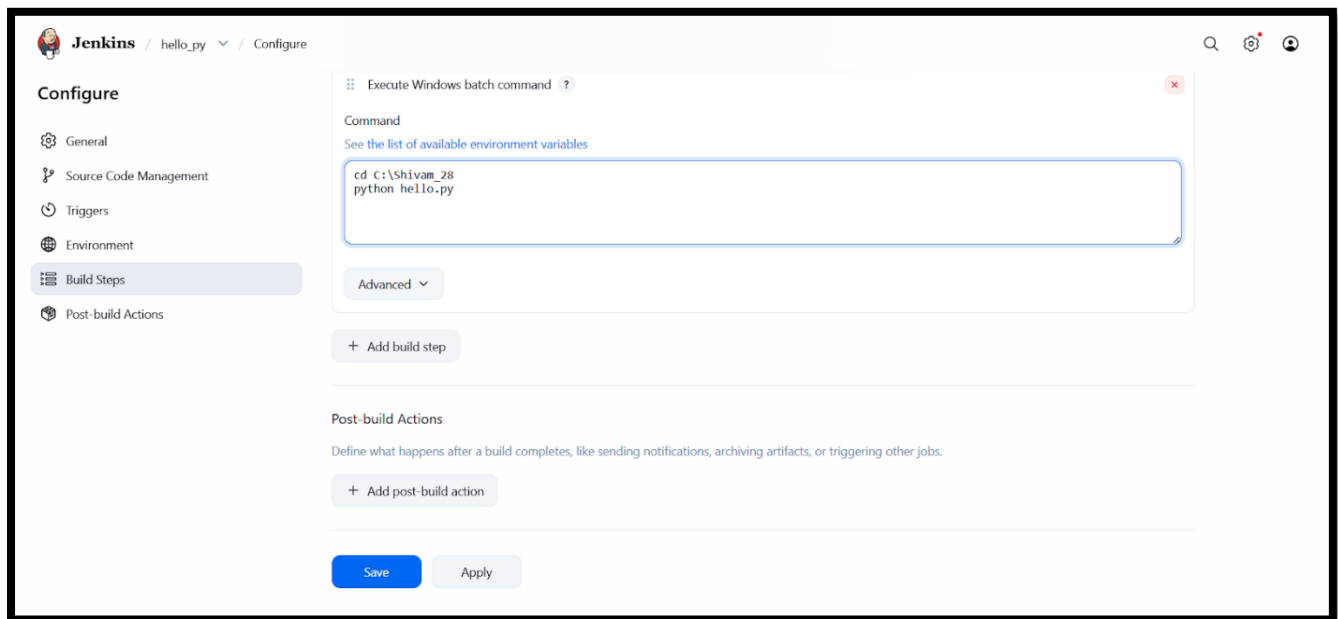
Experiment 5:

1. Python freestyle project – simple python script (implicit)

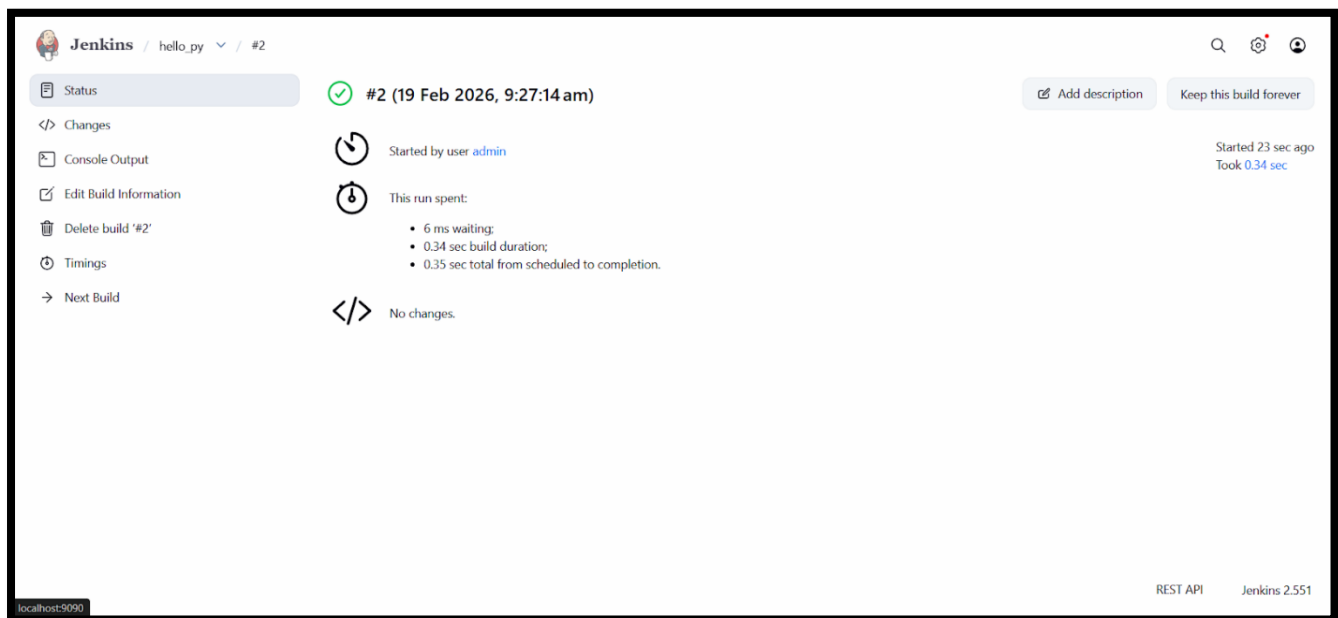
Code (hello.py):

```
print("Hello Jenkins!")
```

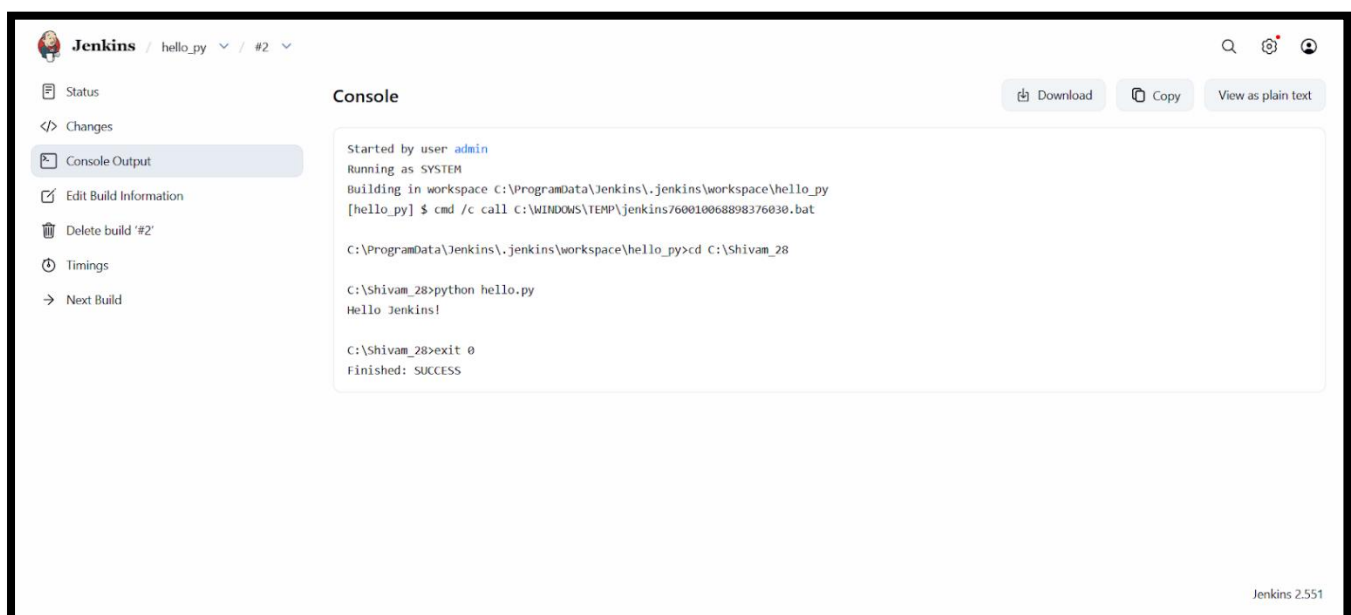




The image shows the Jenkins 'Configure' page for a job named 'hello_py'. The left sidebar contains a menu with options: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'Build Steps' option is selected. The main area is titled 'Execute Windows batch command' and contains a text box with the command: `cd C:\Shivam_28`
`python hello.py`. Below the text box is an 'Advanced' dropdown menu and a '+ Add build step' button. At the bottom, there is a 'Post-build Actions' section with a '+ Add post-build action' button. The 'Save' and 'Apply' buttons are at the bottom right.



The image shows the Jenkins 'Build #2' summary page. The left sidebar contains a menu with options: Status, Changes, Console Output, Edit Build Information, Delete build '#2', Timings, and Next Build. The 'Status' option is selected. The main area shows a green checkmark and the text '#2 (19 Feb 2026, 9:27:14 am)'. Below this, it says 'Started by user admin'. A section titled 'This run spent:' lists the following times: 6 ms waiting, 0.34 sec build duration, and 0.35 sec total from scheduled to completion. At the bottom, it says 'No changes.' The right sidebar contains buttons for 'Add description' and 'Keep this build forever'. The bottom right corner shows 'REST API' and 'Jenkins 2.551'.

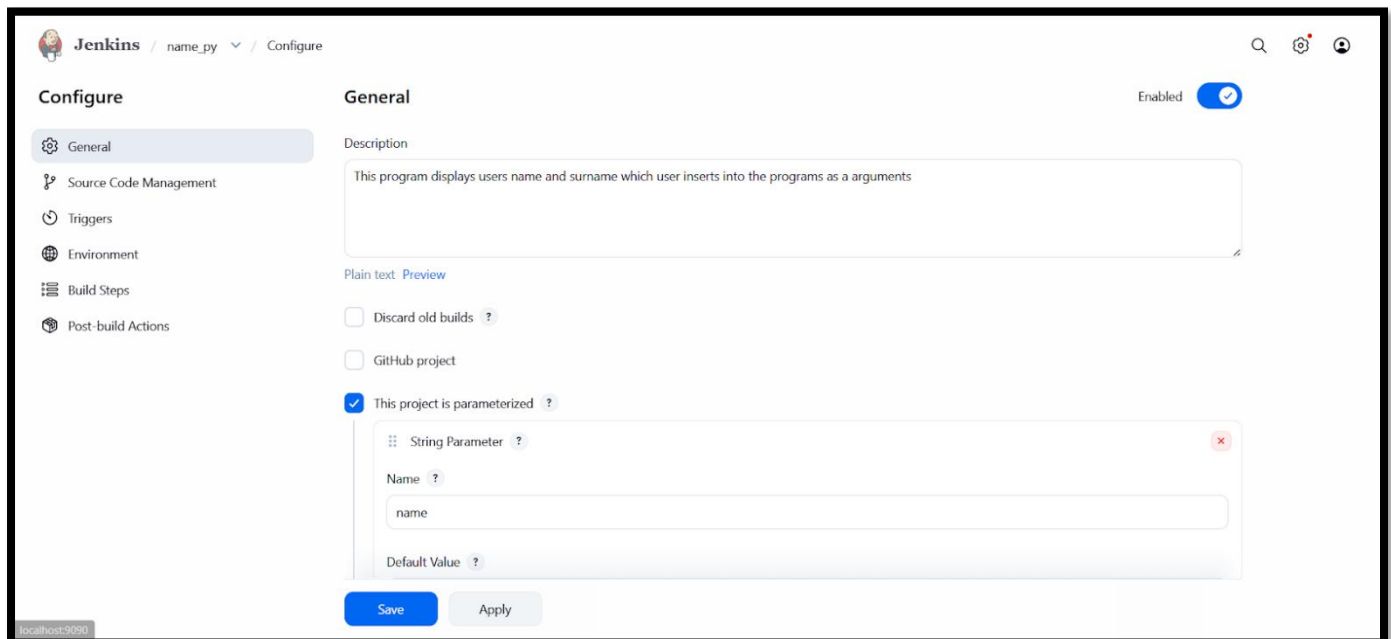
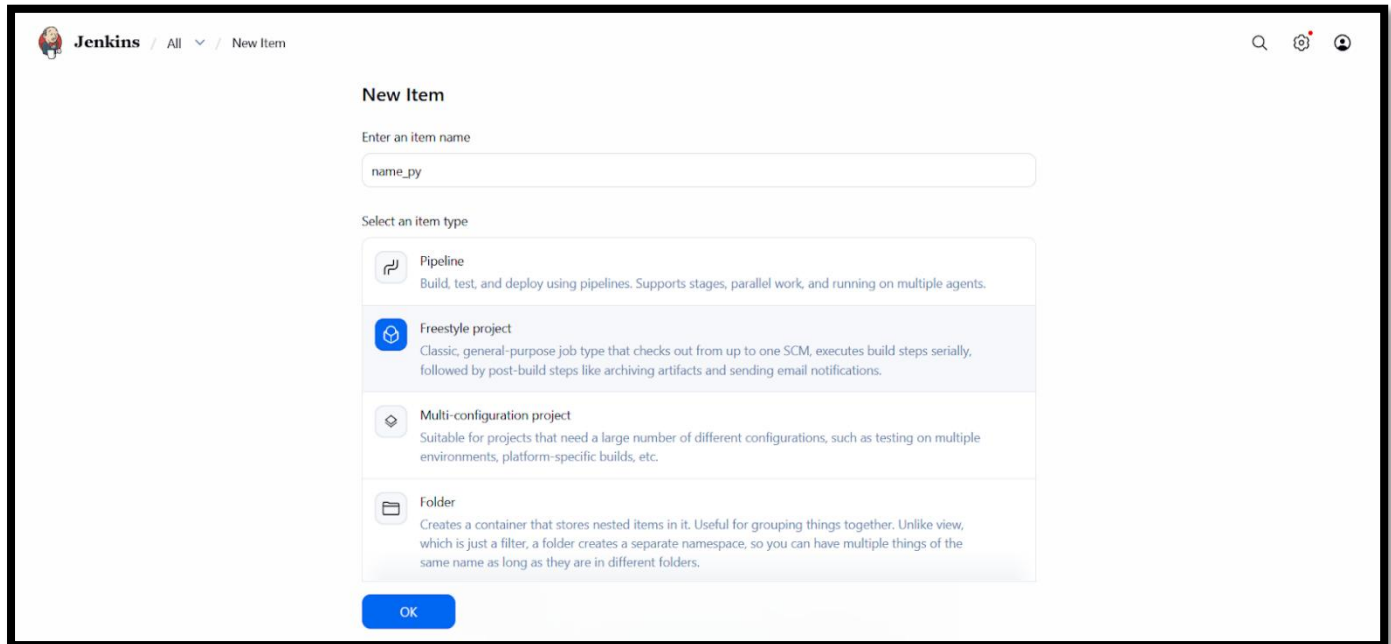


The image shows the Jenkins 'Console Output' page for the 'hello_py' job. The left sidebar contains a menu with options: Status, Changes, Console Output, Edit Build Information, Delete build '#2', Timings, and Next Build. The 'Console Output' option is selected. The main area is titled 'Console' and contains a text box with the following output: `Started by user admin`
`Running as SYSTEM`
`Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\hello_py`
`[hello_py] $ cmd /c call C:\WINDOWS\TEMP\jenkins760010068898376030.bat`
`C:\ProgramData\Jenkins\jenkins\workspace\hello_py>cd C:\Shivam_28`
`C:\Shivam_28>python hello.py`
`Hello Jenkins!`
`C:\Shivam_28>exit 0`
`Finished: SUCCESS`. The right sidebar contains buttons for 'Download', 'Copy', and 'View as plain text'. The bottom right corner shows 'Jenkins 2.551'.

2. Python freestyle project – simple python file (explicit)

Code (name.py):

```
import sys
print("Your name is :", sys.argv[1])
print("Your surname is :", sys.argv[2])
```



Jenkins / name_py / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Name ?
name

Default Value ?

Description ?

Plain text [Preview](#)

☐ Trim the string ?

String Parameter ?

Name ?
surname

Save Apply

Jenkins / name_py / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Execute Windows batch command ?

Command
[See the list of available environment variables](#)

```
cd C:\Shivam_28
python name.py %name% %surname%
```

Advanced ▾

+ Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

+ Add post-build action

Save Apply

localhost:9090

Jenkins / name_py

Project name_py

This build requires parameters:

name
Shivam

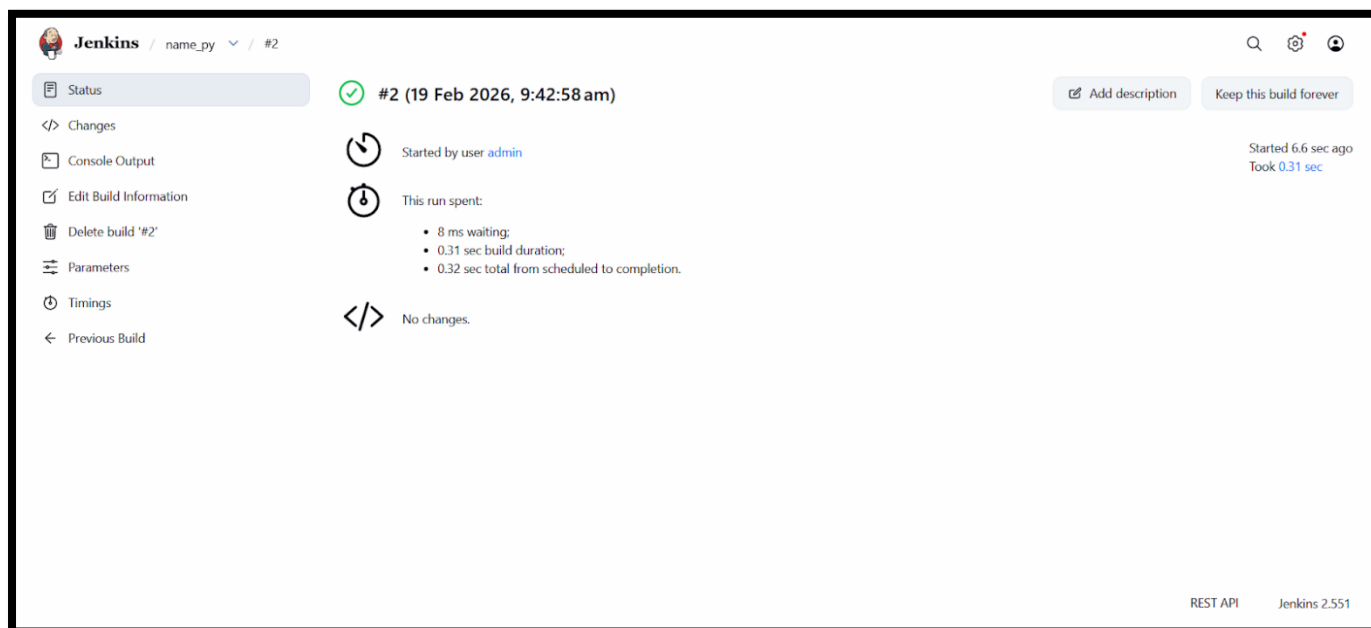
surname
Pimple

Builds > ...

No builds

Build Cancel

Jenkins 2.551



This screenshot shows the Jenkins build status for a job named 'name_py', build #2. The build is successful, indicated by a green checkmark. The build was started by user 'admin' on 19 Feb 2026 at 9:42:58 am. The console output shows that the build took 0.31 seconds and the total time from scheduled to completion was 0.32 seconds. The build was triggered by a change, but no changes were detected. The left sidebar contains links to Status, Changes, Console Output, Edit Build Information, Delete build '#2', Parameters, Timings, and Previous Build. The right sidebar contains links to Add description and Keep this build forever. The bottom right corner shows REST API and Jenkins 2.551.

Jenkins / name_py / #2

Status #2 (19 Feb 2026, 9:42:58 am)

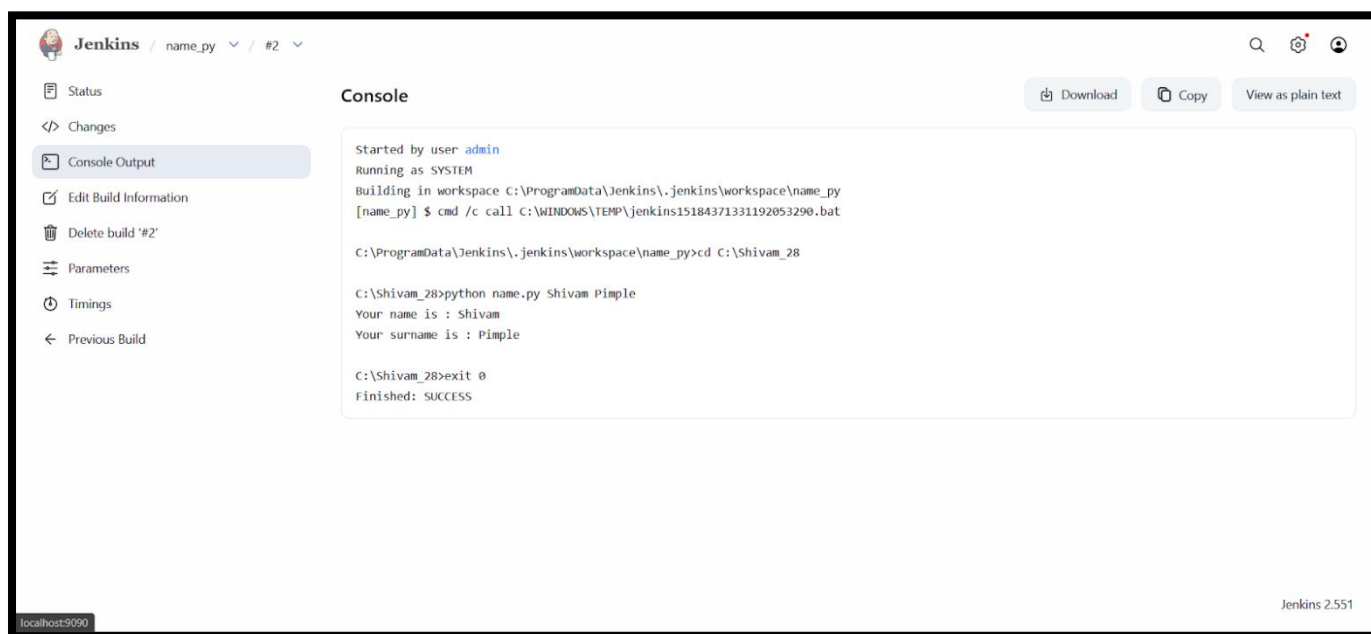
Started by user admin

This run spent:

- 8 ms waiting;
- 0.31 sec build duration;
- 0.32 sec total from scheduled to completion.

No changes.

REST API Jenkins 2.551



This screenshot shows the Jenkins console output for the same build. The console output shows the build was started by user 'admin' and running as SYSTEM. The build is in workspace C:\ProgramData\Jenkins\jenkins\workspace\name_py. The console output shows the command being executed: `[name_py] $ cmd /c call c:\WINDOWS\TEMP\jenkins15184371331192853290.bat`. The output shows the directory being changed to C:\Shivam_28 and the command being executed: `C:\Shivam_28>python name.py Shivam Pimple`. The output shows the name and surname being printed: `Your name is : Shivam` and `Your surname is : Pimple`. The command ends with `C:\Shivam_28>exit 0` and the build finishes successfully: `Finished: SUCCESS`. The left sidebar contains links to Status, Changes, Console Output, Edit Build Information, Delete build '#2', Parameters, Timings, and Previous Build. The right sidebar contains links to Download, Copy, and View as plain text. The bottom left corner shows localhost:9090 and the bottom right corner shows Jenkins 2.551.

Jenkins / name_py / #2

Console

Started by user admin

Running as SYSTEM

Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\name_py

[name_py] \$ cmd /c call c:\WINDOWS\TEMP\jenkins15184371331192853290.bat

C:\ProgramData\Jenkins\jenkins\workspace\name_py>cd C:\Shivam_28

C:\Shivam_28>python name.py Shivam Pimple

Your name is : Shivam

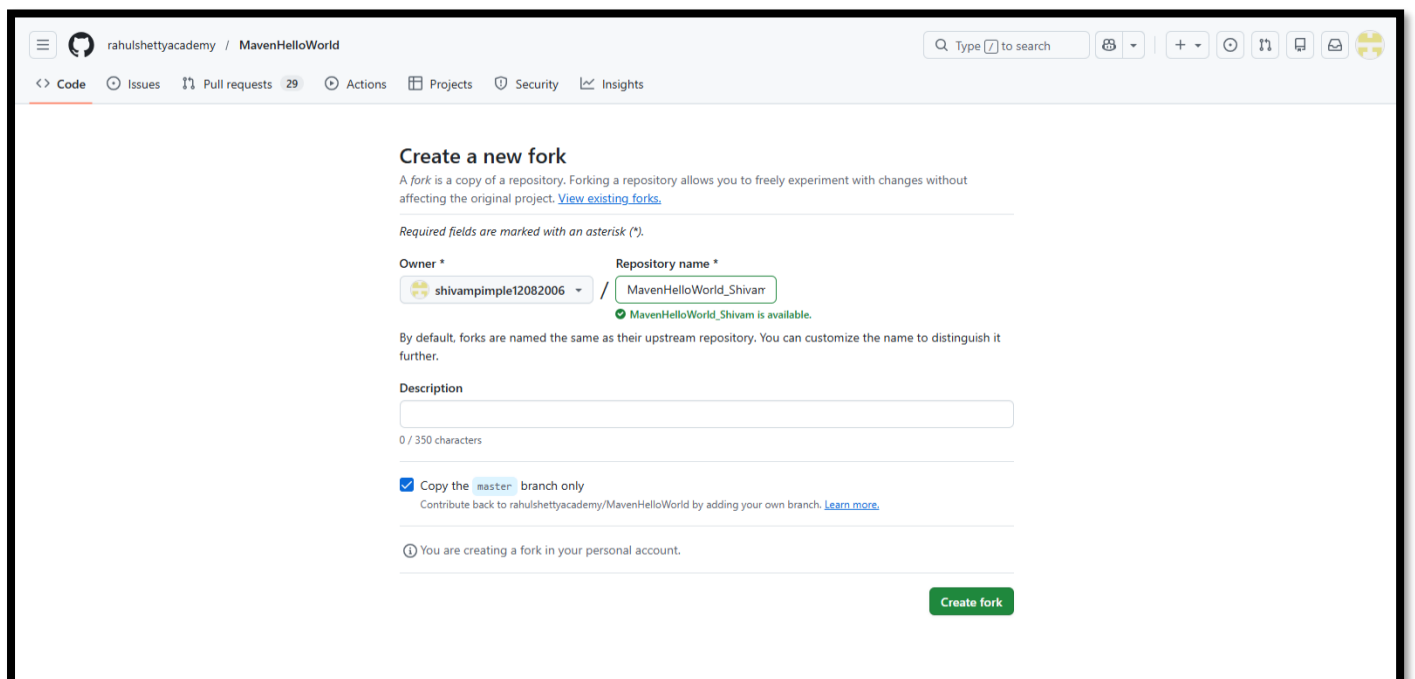
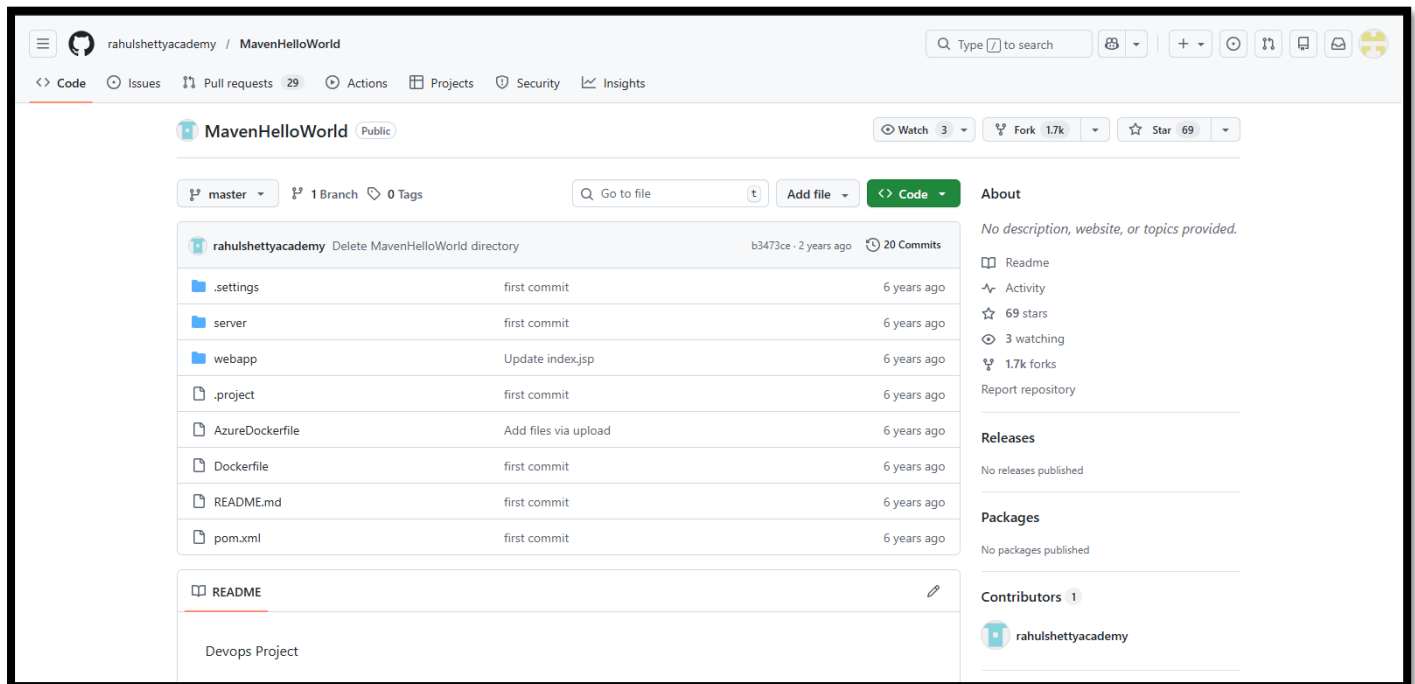
Your surname is : Pimple

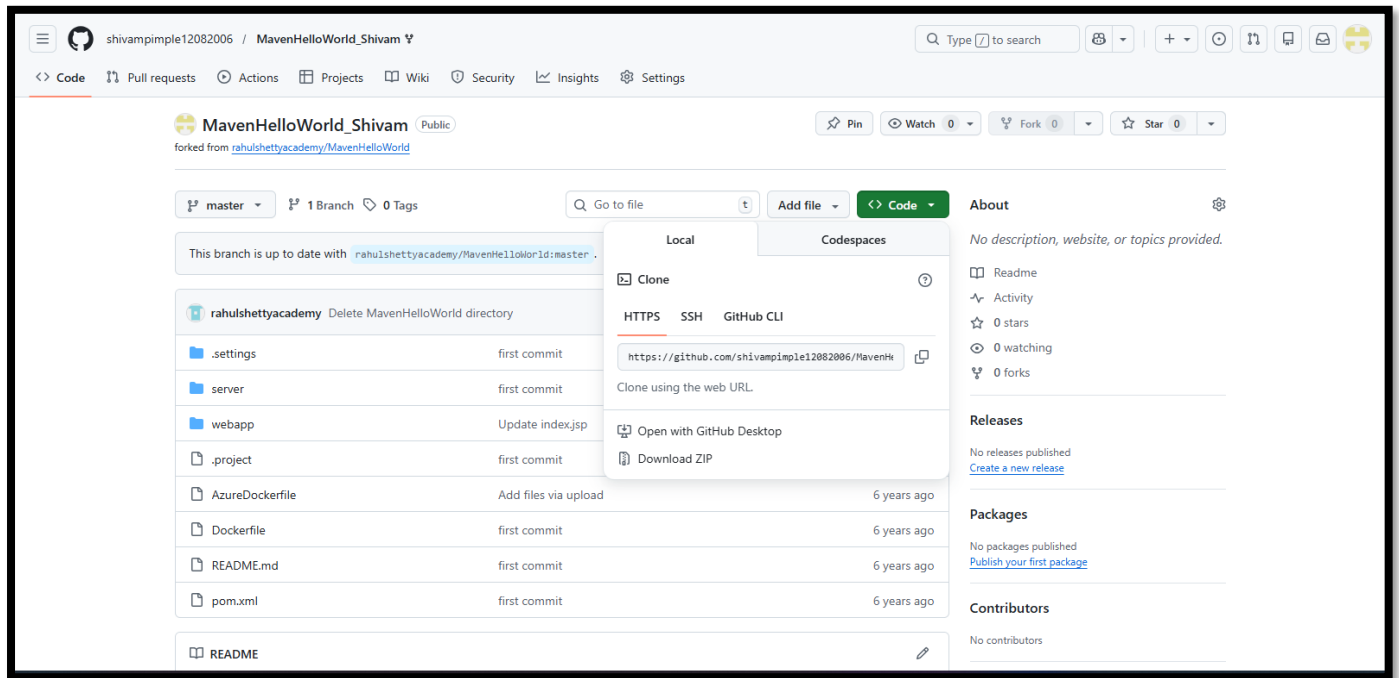
C:\Shivam_28>exit 0

Finished: SUCCESS

localhost:9090 Jenkins 2.551

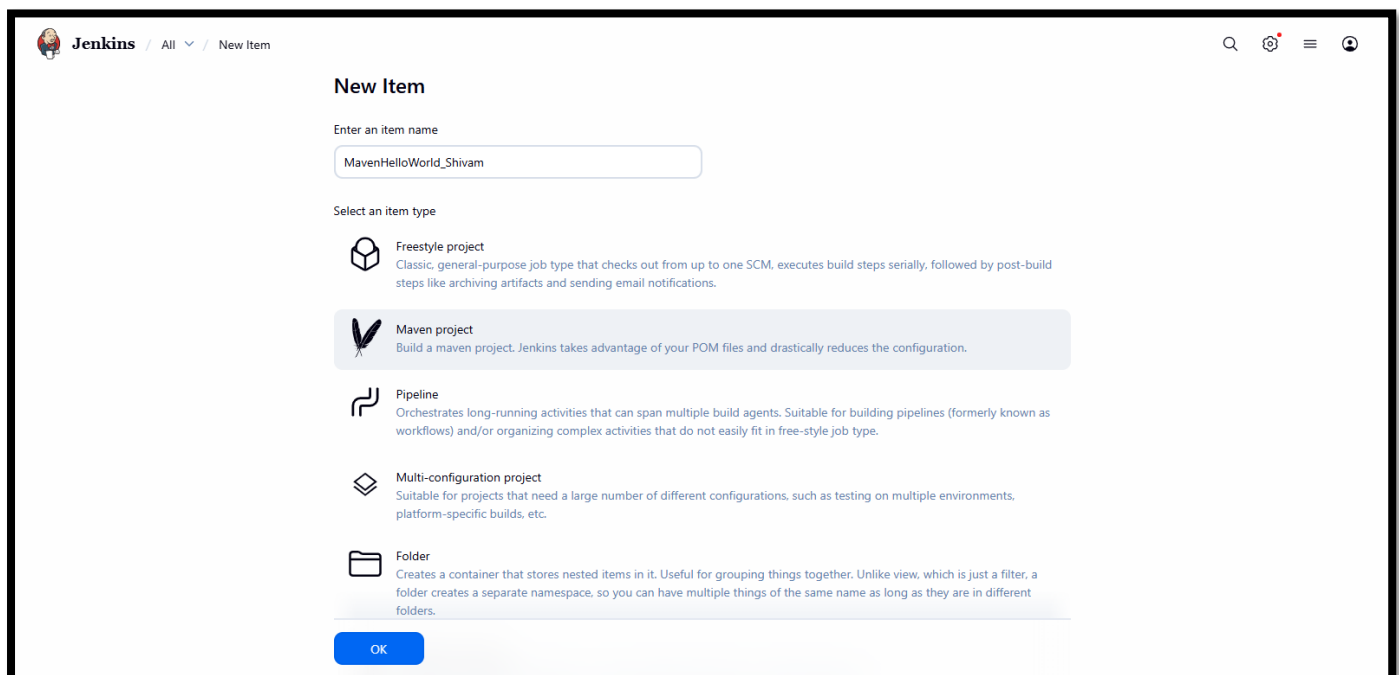
3. Maven freestyle project – Fork repository and Build goals, verify creation of jar/war

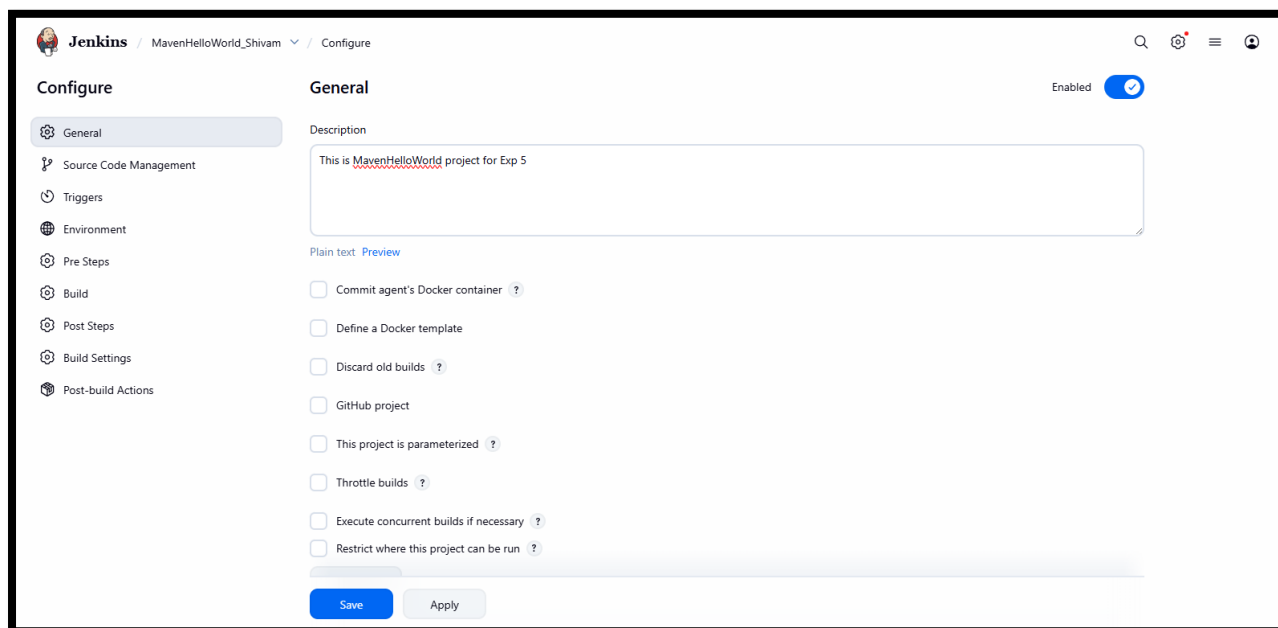




- **GitHub repository link:**

https://github.com/shivampimple12082006/MavenHelloWorld_Shivam.git





Jenkins / MavenHelloWorld_Shivam / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Pre Steps
- Build
- Post Steps
- Build Settings
- Post-build Actions

General

Description

This is MavenHelloWorld project for Exp 5

Plain text [Preview](#)

☐ Commit agent's Docker container ?

☐ Define a Docker template

☐ Discard old builds ?

☐ GitHub project

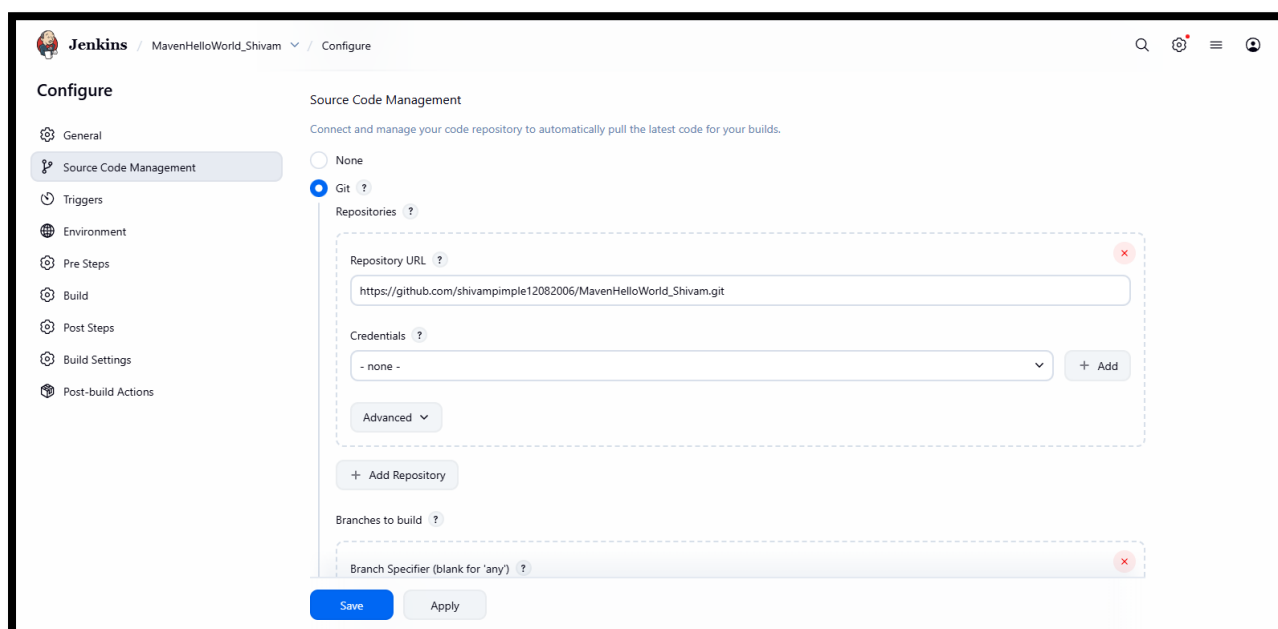
☐ This project is parameterized ?

☐ Throttle builds ?

☐ Execute concurrent builds if necessary ?

☐ Restrict where this project can be run ?

[Save](#) [Apply](#)



Jenkins / MavenHelloWorld_Shivam / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Pre Steps
- Build
- Post Steps
- Build Settings
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ [Git ?](#)

[Repositories ?](#)

Repository URL ?

https://github.com/shivampimple12082006/MavenHelloWorld_Shivam.git

Credentials ?

- none - [+ Add](#)

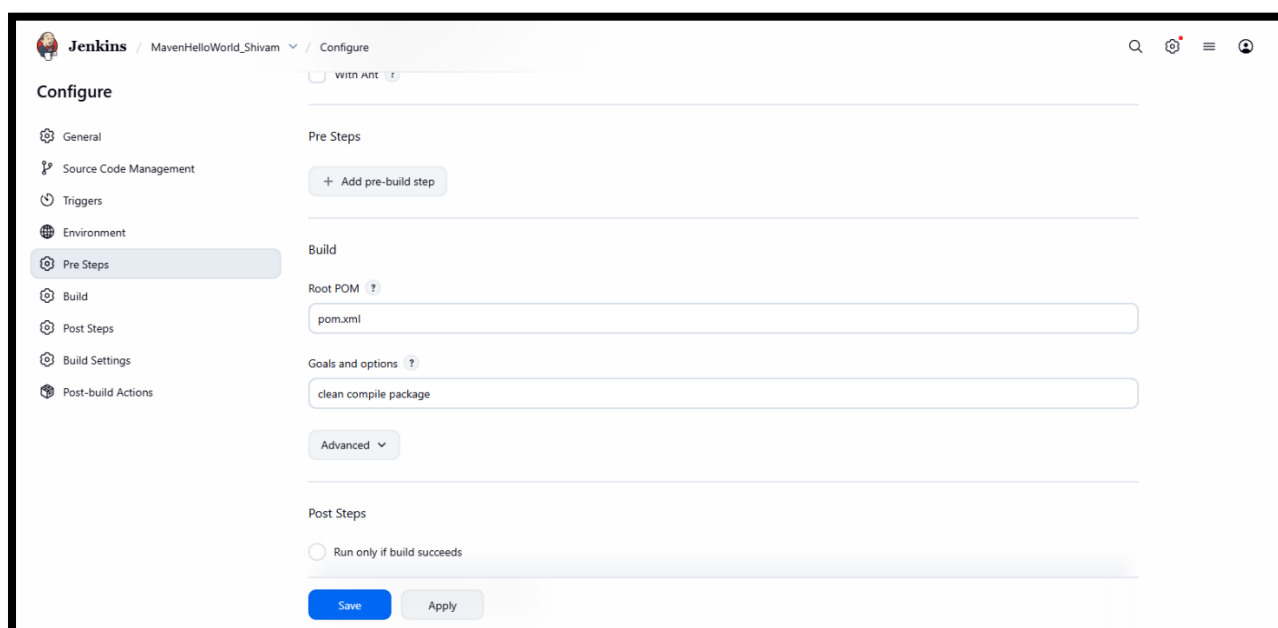
[Advanced](#) ▾

[+ Add Repository](#)

[Branches to build ?](#)

Branch Specifier (blank for 'any') ?

[Save](#) [Apply](#)



Jenkins / MavenHelloWorld_Shivam / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Pre Steps
- Build
- Post Steps
- Build Settings
- Post-build Actions

☐ with ANT ?

Pre Steps

[+ Add pre-build step](#)

Build

Root POM ?

[pom.xml](#)

Goals and options ?

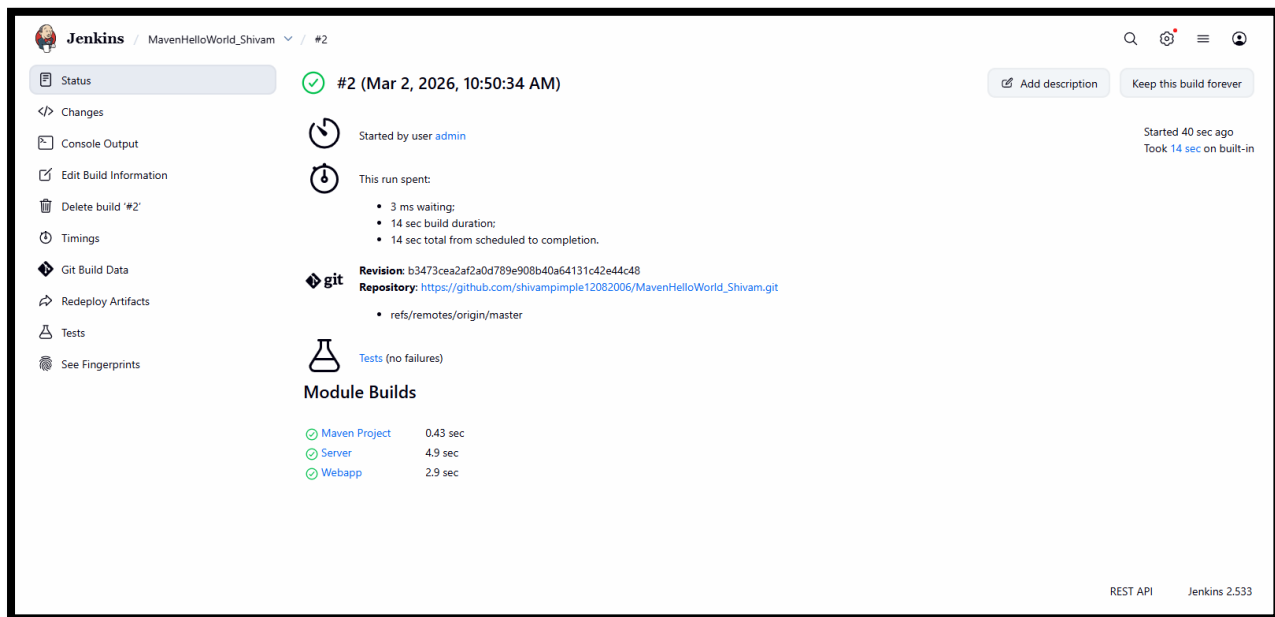
[clean compile package](#)

[Advanced](#) ▾

Post Steps

☐ Run only if build succeeds

[Save](#) [Apply](#)



This screenshot shows the Jenkins build status for a project named 'MavenHelloWorld_Shivam'. The build is labeled '#2' and was completed on March 2, 2026, at 10:50:34 AM. The build status is 'Success' (indicated by a green checkmark). The console output shows that the build started by user 'admin' and took 14 seconds to complete. The build process included a 'git' step to fetch changes from a repository and a 'Tests' step which passed with no failures. The build also includes a 'Module Builds' section showing the duration of each module: Maven Project (0.43 sec), Server (4.9 sec), and Webapp (2.9 sec). The Jenkins REST API endpoint is listed as 'Jenkins 2.533'.

Jenkins / MavenHelloWorld_Shivam / #2

Status **#2** (Mar 2, 2026, 10:50:34 AM)

Started by user **admin**

This run spent:

- 3 ms waiting:
- 14 sec build duration:
- 14 sec total from scheduled to completion.

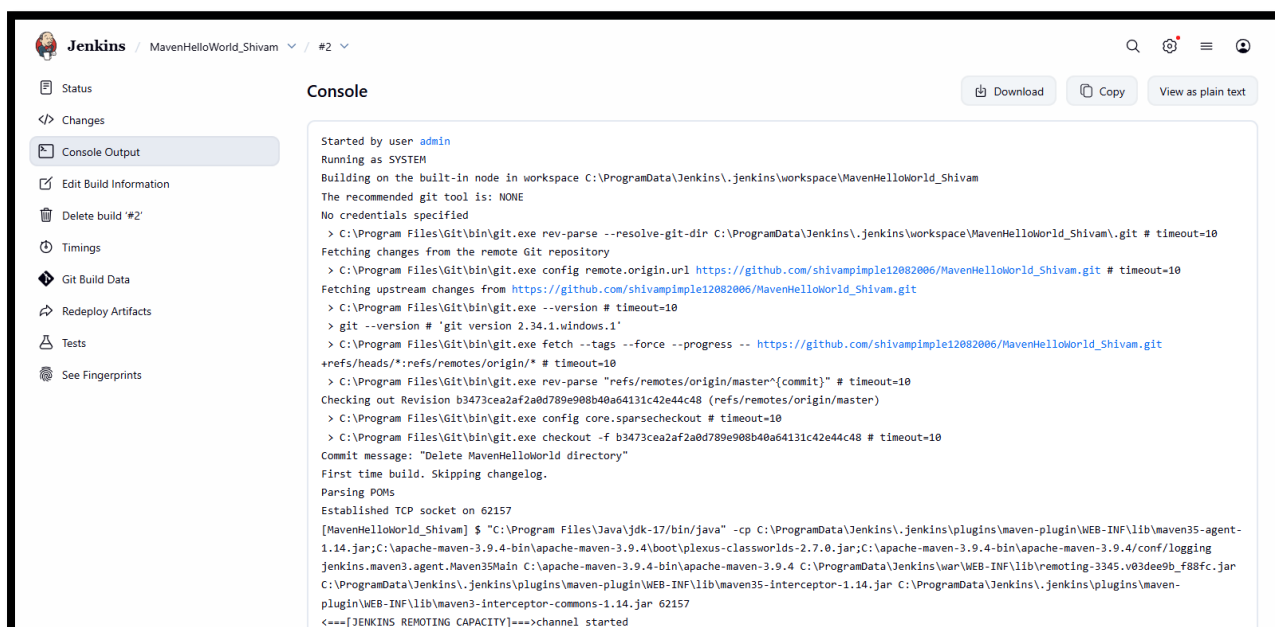
git Revision: b3473cea2af2a0d789e908b40a64131c42e44c48
Repository: https://github.com/shivampimple12082006/MavenHelloWorld_Shivam.git

Tests (no failures)

Module Builds

Module	Duration
Maven Project	0.43 sec
Server	4.9 sec
Webapp	2.9 sec

REST API Jenkins 2.533



This screenshot shows the console output of the Jenkins build. The output starts with 'Started by user admin' and 'Running as SYSTEM'. It then shows the build process for the 'MavenHelloWorld_Shivam' project, including the recommended git tool, fetching changes from the remote repository, and checking out the revision b3473cea2af2a0d789e908b40a64131c42e44c48. The console output also shows the build process for the 'Maven Project', 'Server', and 'Webapp' modules, and the final build status 'BUILD SUCCESS'.

Jenkins / MavenHelloWorld_Shivam / #2

Status Console

Started by user **admin**

Running as SYSTEM

Building on the built-in node in workspace C:\ProgramData\Jenkins\jenkins\workspace\MavenHelloWorld_Shivam

The recommended git tool is: NONE

No credentials specified

> C:\Program Files\Git\bin\git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\jenkins\workspace\MavenHelloWorld_Shivam\git # timeout=10

Fetching changes from the remote Git repository

> C:\Program Files\Git\bin\git.exe config remote.origin.url https://github.com/shivampimple12082006/MavenHelloWorld_Shivam.git # timeout=10

Fetching upstream changes from https://github.com/shivampimple12082006/MavenHelloWorld_Shivam.git

> C:\Program Files\Git\bin\git.exe --version # timeout=10

> git --version # 'git version 2.34.1.windows.1'

> C:\Program Files\Git\bin\git.exe fetch --tags --force --progress -- https://github.com/shivampimple12082006/MavenHelloWorld_Shivam.git +refs/heads/*:refs/remotes/origin/* # timeout=10

> C:\Program Files\Git\bin\git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10

Checking out Revision b3473cea2af2a0d789e908b40a64131c42e44c48 (refs/remotes/origin/master)

> C:\Program Files\Git\bin\git.exe config core.sparsecheckout # timeout=10

> C:\Program Files\Git\bin\git.exe checkout -f b3473cea2af2a0d789e908b40a64131c42e44c48 # timeout=10

Commit message: "Delete MavenHelloWorld directory"

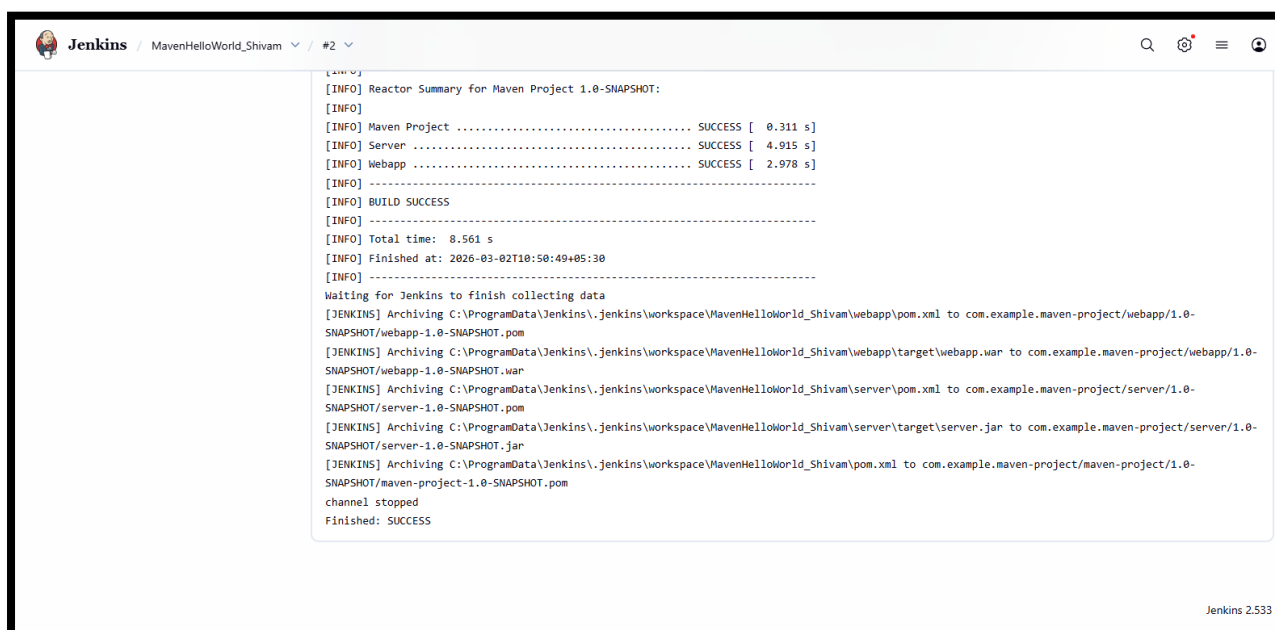
First time build. Skipping changelog.

Parsing POMs

Established TCP socket on 62157

[MavenHelloWorld_Shivam] \$ "C:\Program Files\Java\jdk-17\bin\java" -cp C:\ProgramData\Jenkins\jenkins\plugins\maven-plugin\WEB-INF\lib\maven3-agent-1.14.jar;C:\apache-maven-3.9.4\bin\apache-maven-3.9.4\boot\plexus-classworlds-2.7.0.jar;C:\apache-maven-3.9.4\bin\apache-maven-3.9.4\conf\logging-jenkins-maven3-agent.Maven3Main C:\apache-maven-3.9.4\bin\apache-maven-3.9.4 C:\ProgramData\Jenkins\war\WEB-INF\lib\remoting-3345.v03dee9b_f88fc.jar C:\ProgramData\Jenkins\jenkins\plugins\maven-plugin\WEB-INF\lib\maven3-interceptor-1.14.jar C:\ProgramData\Jenkins\jenkins\plugins\maven-plugin\WEB-INF\lib\maven3-interceptor-commons-1.14.jar 62157

<===[JENKINS REMOTING CAPACITY]===>channel started



This screenshot shows the console output of the Jenkins build, specifically the Maven build process. The output shows the Maven Project, Server, and Webapp modules being built successfully. The build process includes archiving the project files and the final build status 'BUILD SUCCESS'.

Jenkins / MavenHelloWorld_Shivam / #2

[INFO] Reactor Summary for Maven Project 1.0-SNAPSHOT:

[INFO]

[INFO] Maven Project SUCCESS [0.311 s]

[INFO] Server SUCCESS [4.915 s]

[INFO] Webapp SUCCESS [2.978 s]

[INFO] -----

[INFO] BUILD SUCCESS

[INFO] -----

[INFO] Total time: 8.561 s

[INFO] Finished at: 2026-03-02T10:50:49+05:30

[INFO] -----

Waiting for Jenkins to finish collecting data

[JENKINS] Archiving C:\ProgramData\Jenkins\jenkins\workspace\MavenHelloWorld_Shivam\webapp\pom.xml to com.example.maven-project\webapp/1.0-SNAPSHOT/webapp-1.0-SNAPSHOT.pom

[JENKINS] Archiving C:\ProgramData\Jenkins\jenkins\workspace\MavenHelloWorld_Shivam\webapp\target\webapp.war to com.example.maven-project\webapp/1.0-SNAPSHOT/webapp-1.0-SNAPSHOT.war

[JENKINS] Archiving C:\ProgramData\Jenkins\jenkins\workspace\MavenHelloWorld_Shivam\server\pom.xml to com.example.maven-project/server/1.0-SNAPSHOT/server-1.0-SNAPSHOT.pom

[JENKINS] Archiving C:\ProgramData\Jenkins\jenkins\workspace\MavenHelloWorld_Shivam\server\target\server.jar to com.example.maven-project/server/1.0-SNAPSHOT/server-1.0-SNAPSHOT.jar

[JENKINS] Archiving C:\ProgramData\Jenkins\jenkins\workspace\MavenHelloWorld_Shivam\pom.xml to com.example.maven-project/maven-project/1.0-SNAPSHOT/maven-project-1.0-SNAPSHOT.pom

channel stopped

Finished: SUCCESS

Jenkins 2.533

```
C:\Users\MOAD>cd C:\ProgramData\Jenkins\.jenkins\workspace\MavenHelloWorld_Shivam\webapp\target

C:\ProgramData\Jenkins\.jenkins\workspace\MavenHelloWorld_Shivam\webapp\target>dir
Volume in drive C has no label.
Volume Serial Number is 50D5-6323

Directory of C:\ProgramData\Jenkins\.jenkins\workspace\MavenHelloWorld_Shivam\webapp\target

02-03-2026  10:50    <DIR>          .
02-03-2026  10:50    <DIR>          ..
02-03-2026  10:50    <DIR>          maven-archiver
02-03-2026  10:50    <DIR>          surefire
02-03-2026  10:50    <DIR>          webapp
02-03-2026  10:50                2,053 webapp.war
               1 File(s)                2,053 bytes
               5 Dir(s) 122,137,292,800 bytes free

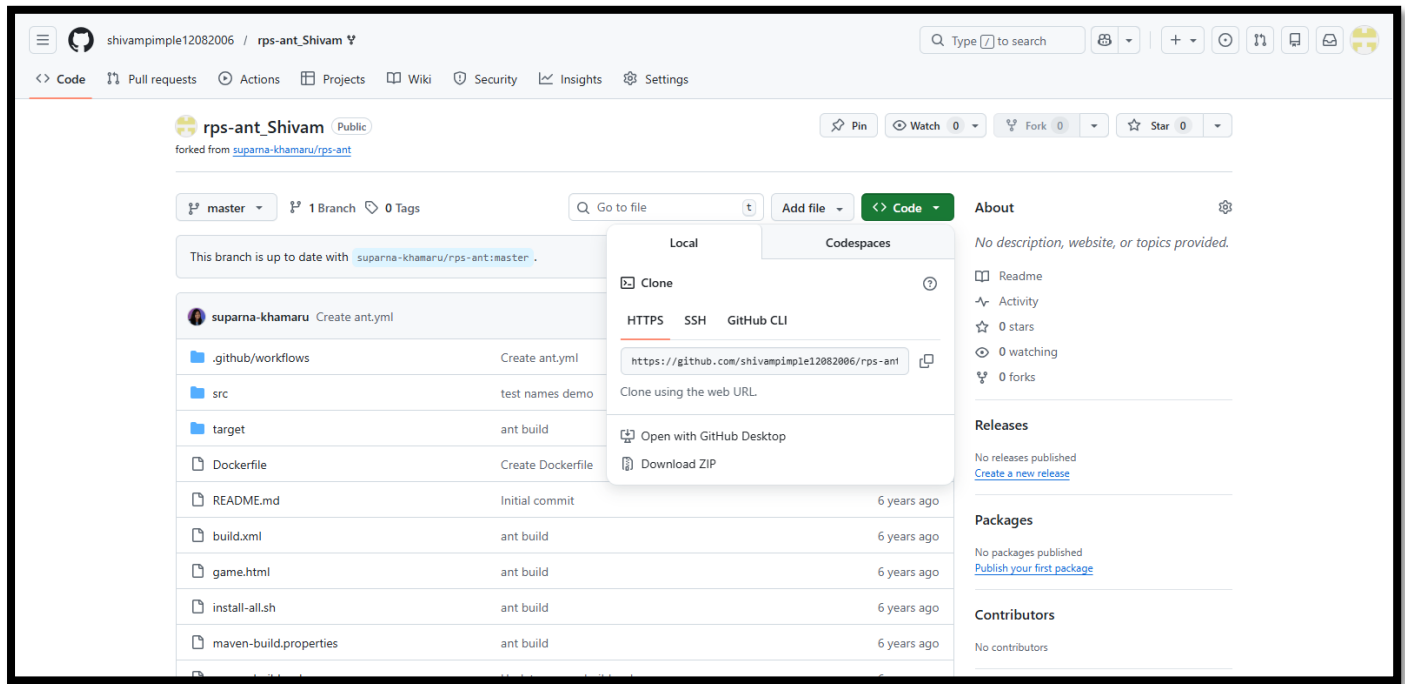
C:\ProgramData\Jenkins\.jenkins\workspace\MavenHelloWorld_Shivam\webapp\target>
```

4. Ant freestyle project - Fork repository and Build goals, verify creation of jar/war

The screenshot shows the GitHub repository page for 'rps-ant' by user 'suparna-khamaru'. The repository is public and has 0 watches, 672 forks, and 3 stars. The file list includes:

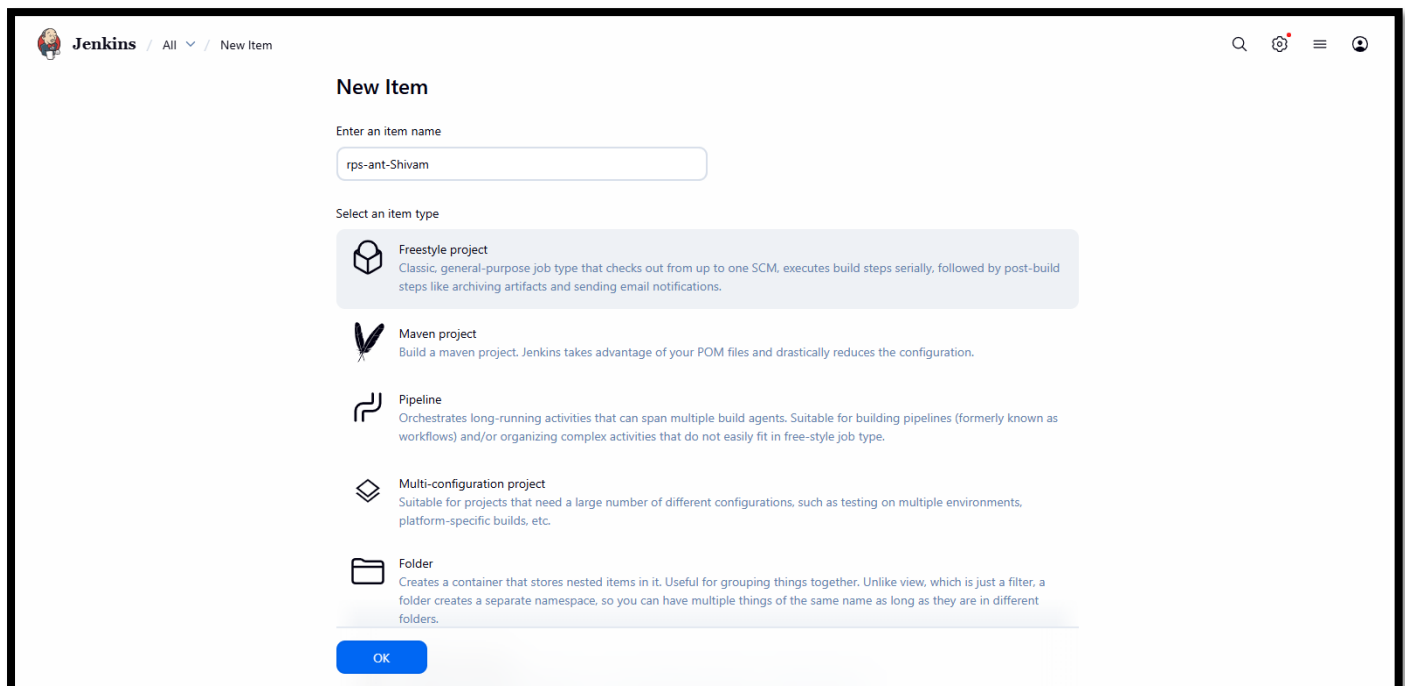
File	Commit	Time
suparna-khamaru Create ant.yml	02f88a3	5 years ago
.github/workflows	Create ant.yml	5 years ago
src	test names demo	6 years ago
target	ant build	6 years ago
Dockerfile	Create Dockerfile	6 years ago
README.md	Initial commit	6 years ago
build.xml	ant build	6 years ago
game.html	ant build	6 years ago
install-all.sh	ant build	6 years ago
maven-build.properties	ant build	6 years ago
maven-build.xml	Update maven-build.xml	6 years ago
pom.xml	ant build	6 years ago
scripted-pipeline	ant build	6 years ago

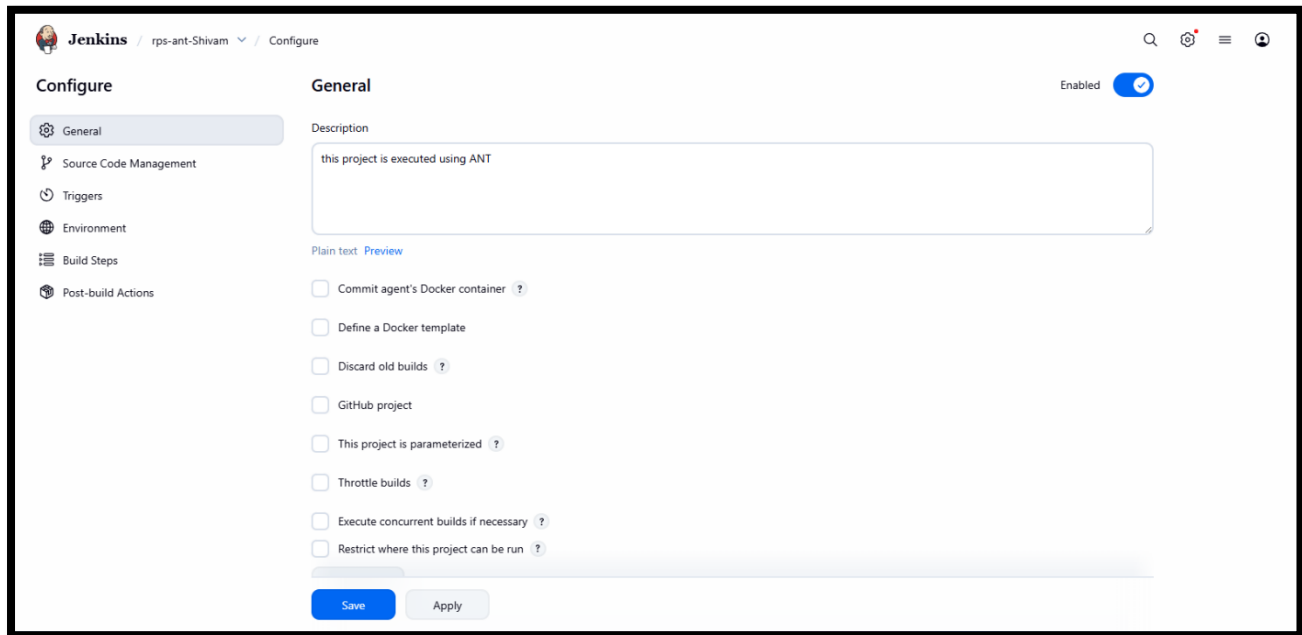
The screenshot shows the 'Create a new fork' form on GitHub. The form is for creating a fork of the repository 'rps-ant' by 'suparna-khamaru'. The 'Owner' is set to 'shivampimple12082006' and the 'Repository name' is 'rps-ant_Shivam'. The 'Description' field is empty. The 'Copy the master branch only' checkbox is checked. The 'Create fork' button is at the bottom right.



- **GitHub repository link:**

https://github.com/shivampimple12082006/rps-ant_Shivam.git





Jenkins / rps-ant-Shivam / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

General Enabled

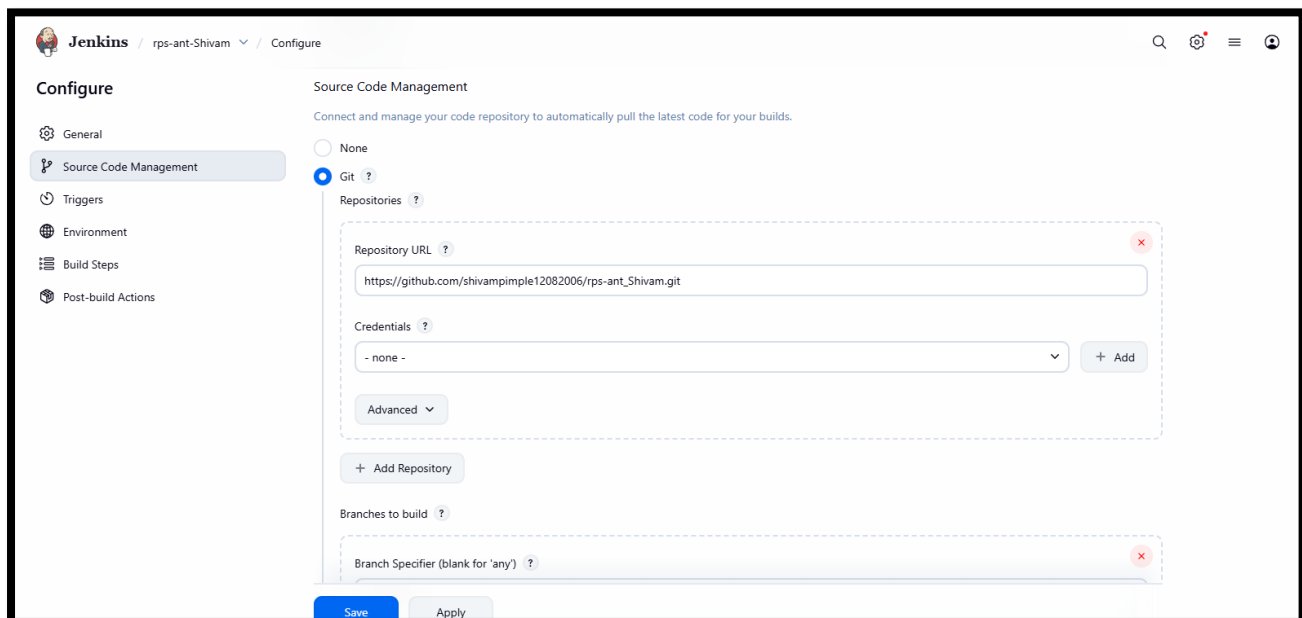
Description

this project is executed using ANT

Plain text [Preview](#)

- ☐ Commit agent's Docker container ?
- ☐ Define a Docker template
- ☐ Discard old builds ?
- ☐ GitHub project
- ☐ This project is parameterized ?
- ☐ Throttle builds ?
- ☐ Execute concurrent builds if necessary ?
- ☐ Restrict where this project can be run ?

[Save](#) [Apply](#)



Jenkins / rps-ant-Shivam / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ [Git ?](#)

[Repositories ?](#)

Repository URL ?

https://github.com/shivampimple12082006/rps-ant_Shivam.git

Credentials ?

- none - [+ Add](#)

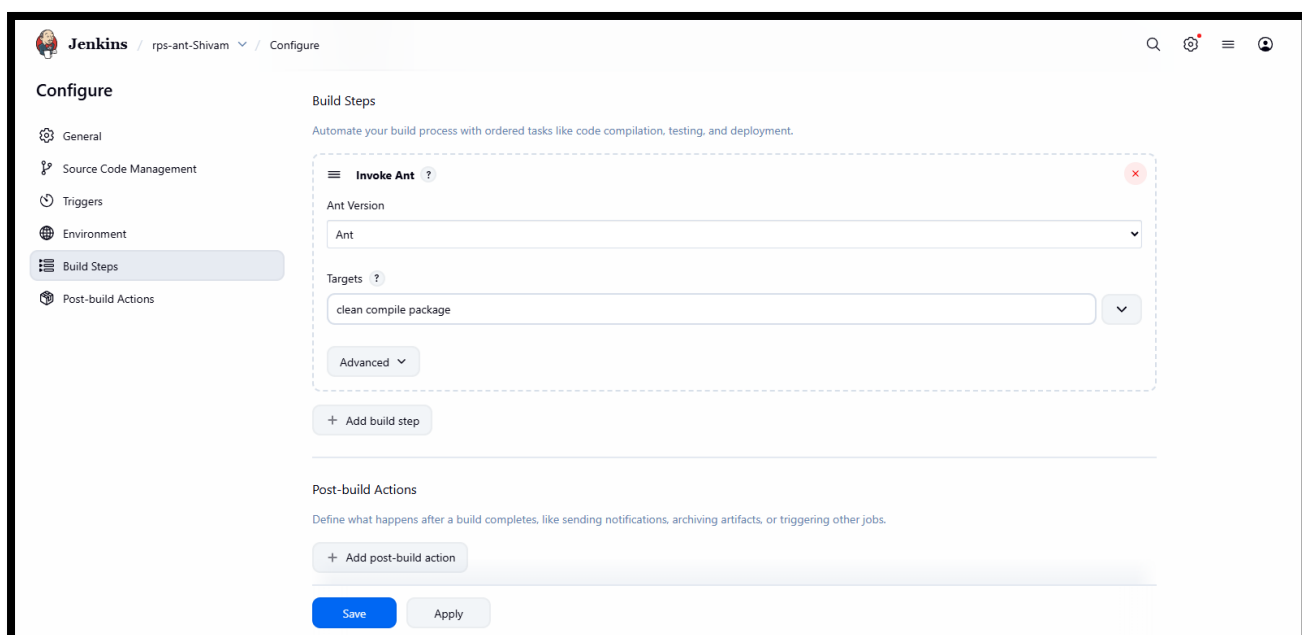
[Advanced](#) ▾

[+ Add Repository](#)

[Branches to build ?](#)

Branch Specifier (blank for 'any') ?

[Save](#) [Apply](#)



Jenkins / rps-ant-Shivam / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

[Invoke Ant ?](#)

Ant Version

Ant

Targets ?

clean compile package

[Advanced](#) ▾

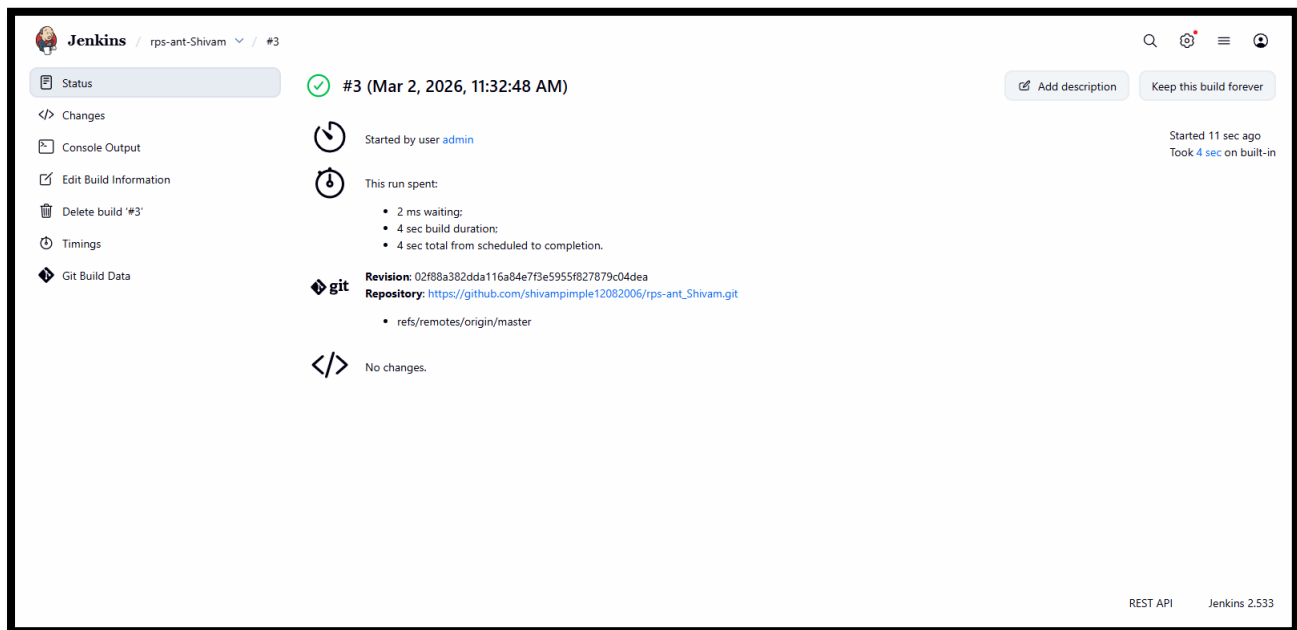
[+ Add build step](#)

Post-build Actions

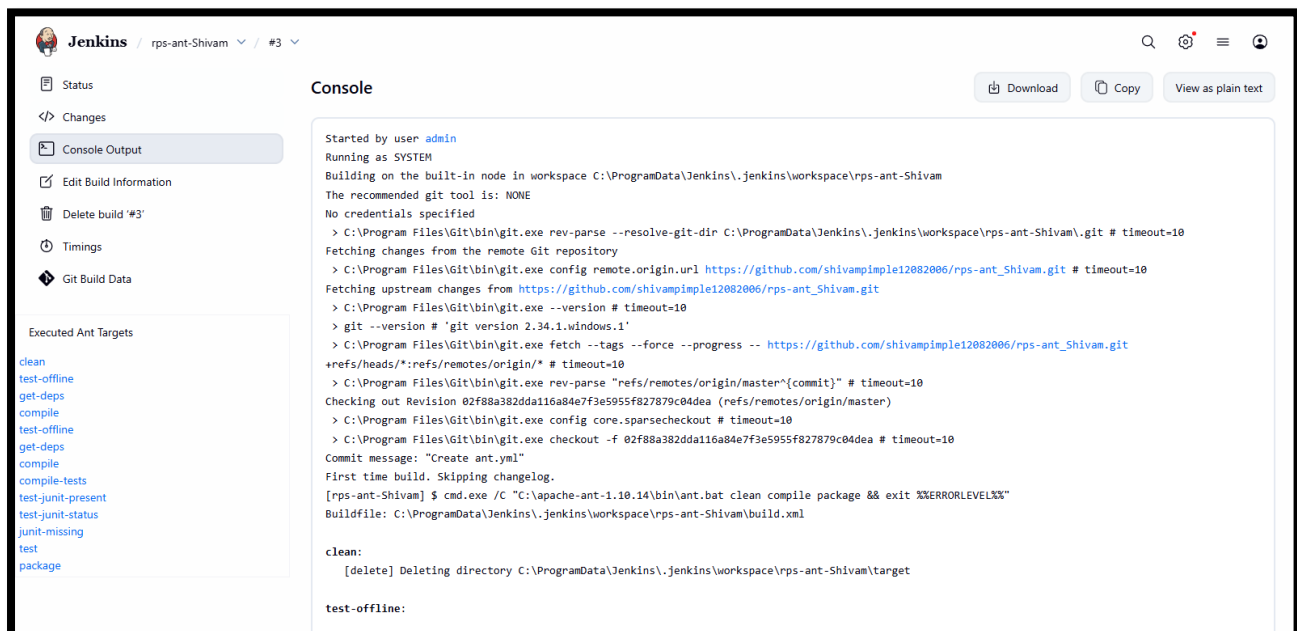
Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

[+ Add post-build action](#)

[Save](#) [Apply](#)



This screenshot shows the Jenkins build status for a job named 'rps-ant-Shivam'. The build is labeled '#3' and was completed on March 2, 2026, at 11:32:48 AM. The status is 'Success' (green checkmark). The build was started by user 'admin' and took 4 seconds to complete on the built-in executor. The console output shows that the build was triggered by a user, and the run spent 2 ms waiting, 4 sec build duration, and 4 sec total from scheduled to completion. The build is using the 'git' plugin to fetch changes from the repository 'https://github.com/shivampimple12082006/rps-ant-Shivam.git'. The revision is '02f88a382dda116a84e7f3e5955f827879c04dea' and the repository is 'https://github.com/shivampimple12082006/rps-ant-Shivam.git'. The build is using the 'refs/remotes/origin/master' branch. The build is using the 'No changes' option. The build is using the 'REST API' and 'Jenkins 2.533'.



This screenshot shows the Jenkins console output for the build. The console output shows the build process starting with 'Started by user admin' and 'Running as SYSTEM'. The build is running on the built-in node in workspace 'C:\ProgramData\Jenkins\jenkins\workspace\rps-ant-Shivam'. The recommended git tool is 'NONE'. The build is using the 'git' plugin to fetch changes from the repository 'https://github.com/shivampimple12082006/rps-ant-Shivam.git'. The build is using the 'refs/remotes/origin/master' branch. The build is using the 'No changes' option. The build is using the 'REST API' and 'Jenkins 2.533'.



This screenshot shows the Jenkins build summary for the build. The build is labeled '#3' and was completed on March 2, 2026, at 11:32:48 AM. The status is 'Success' (green checkmark). The build was started by user 'admin' and took 4 seconds to complete on the built-in executor. The build is using the 'git' plugin to fetch changes from the repository 'https://github.com/shivampimple12082006/rps-ant-Shivam.git'. The revision is '02f88a382dda116a84e7f3e5955f827879c04dea' and the repository is 'https://github.com/shivampimple12082006/rps-ant-Shivam.git'. The build is using the 'refs/remotes/origin/master' branch. The build is using the 'No changes' option. The build is using the 'REST API' and 'Jenkins 2.533'.

```
C:\>cd C:\ProgramData\Jenkins\.jenkins\workspace\rps-ant-Shivam
C:\ProgramData\Jenkins\.jenkins\workspace\rps-ant-Shivam>dir
Volume in drive C has no label.
Volume Serial Number is 50D5-6323

Directory of C:\ProgramData\Jenkins\.jenkins\workspace\rps-ant-Shivam

02-03-2026  11:32    <DIR>          .
02-03-2026  11:28    <DIR>          ..
02-03-2026  11:28    <DIR>          .github
02-03-2026  11:28             992 build.xml
02-03-2026  11:28             804 Dockerfile
02-03-2026  11:28          5,526 game.html
02-03-2026  11:28          1,283 install-all.sh
02-03-2026  11:28             834 maven-build.properties
02-03-2026  11:28          11,136 maven-build.xml
02-03-2026  11:28          2,210 pom.xml
02-03-2026  11:28              9 README.md
02-03-2026  11:28          1,050 scripted-pipeline
02-03-2026  11:28    <DIR>          src
02-03-2026  11:32    <DIR>          target
                9 File(s)          23,844 bytes
                5 Dir(s)  122,032,775,168 bytes free

C:\ProgramData\Jenkins\.jenkins\workspace\rps-ant-Shivam>
```